CSET CENTER *for* SECURITY *and* EMERGING TECHNOLOGY

Summary of "AI for Military Decision-Making: Harnessing the Advantages and Avoiding the Risks"

The integration of artificial intelligence into military operations has become a significant focus for armed forces globally. Military commanders are interested in AI's potential to improve decision-making, especially at the operational level of war, where they must integrate a lot of information quickly to make life-and-death decisions.

Interest in deploying AI-enabled decision support systems (DSS) can be found in most major militaries. Global statements about AI-enabled DSS range from the **circumspect** **("Providing high-level strategy decisions is currently beyond the state-of-the-art of AI, but there is ongoing work to broaden the applicability of AI techniques")** to the **fanciful ("eliminat[e] uncertainty in situation assessment, ensuring near-absolute impartiality in political and military decisions").**

The enthusiasm for AI-DSS must be balanced with an understanding of their capabilities and limitations to ensure appropriate and effective deployment. This paper reviews recently proposed uses of AI-enabled DSS, provides a simplified framework for considering AI-DSS capabilities and limitations, and recommends practical risk mitigations that commanders might employ when operating with an AI-enabled DSS. Our framework for considering AI-DSS intended for operational decision-making emphasizes considerations around system **scoping**, **data**, and **human-machine interaction**.

Based on our analysis, we recommend the following risk mitigation strategies when military commanders use AI-DSS:

1. Set context- and risk-based criteria for deployment: Commanders should set the time, place, and context for deploying and authorizing DSS use and prepare forces to adapt software settings as conditions and risk tolerance change.

2. Train and qualify AI-enabled DSS operators: Operators should be thoroughly trained on DSS capabilities and limitations. Those involved in lethal operations should undergo examinations for official qualifications appropriate to their role in operating the system.

3. Establish a continuous certification cycle: Units leveraging AI-DSS should be regularly certified to reduce the risk of inappropriately deploying or operating the system. Sharing performance metrics with data scientists, operations analysts, and experts in continuous tests and evaluations can help validate continued responsible use of AI-DSS and also support technical evolutions.

4. Designate a Responsible AI Safety Officer: Akin to established safety and mishap programs, Responsible AI safety (RAI) officers in military units can serve as local conduits for new information, promoting broad-based AI literacy, reporting AI incidents or mishaps to a higher authority, and mitigating AI-DSS risks.

5. Document incidents and harms: Documenting AI system flaws and user mistakes is essential for avoiding repeat errors and for building trust through transparency. RAIAI safety officers should be responsible for such documentation, akin to mishap reporting processes already in place in the services.

The integration of AI into military decision-making presents both opportunities and challenges. By carefully considering the scope, data quality, and human-machine interaction, and by implementing rigorous training, certification, and safety measures, military organizations can leverage AI more effectively while mitigating potential risks.

For more information:

- Download the report: https://cset.georgetown.edu/publication/ai-for-military-decision-making/
- Contact us: cset@georgetown.edu